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See Figure 1 and Figure 2.

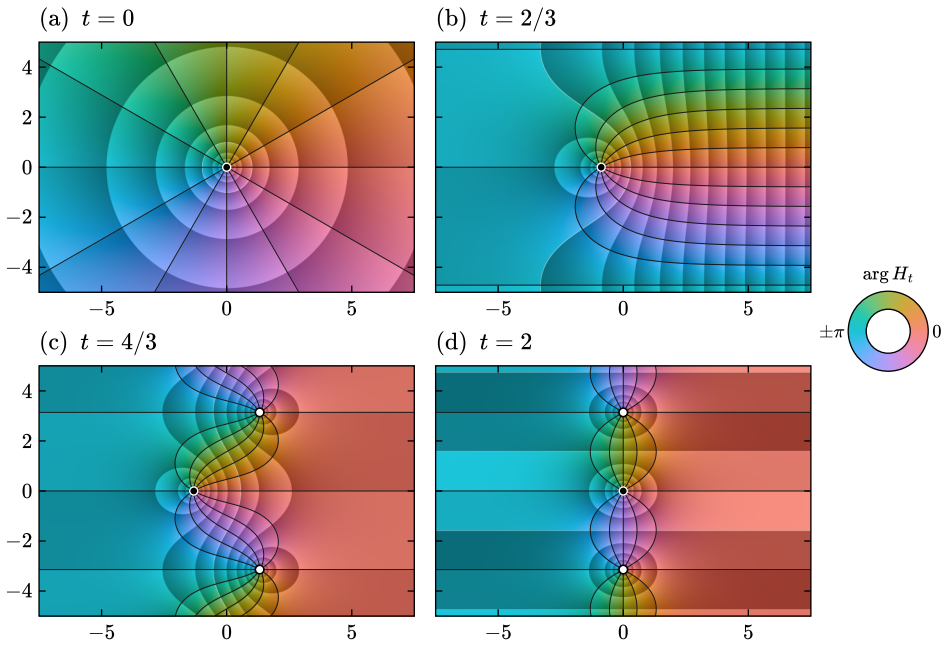


FIGURE 1. Phase portraits of H_t on $[-7.5, 7.5] \times [-5, 5]$ at (a) $t = 0$, (b) $t = 2/3$, (c) $t = 4/3$, and (d) $t = 2$: the color shows $\arg H_t$ (key at right), the thin lines are the level sets $\arg H_t \in \frac{\pi}{6}\mathbb{Z}$, and the shading steps in $|H_t|$ by factors of $e^{2\pi/12} \approx 1.69$; the white line is $|H_t| = 1$. Zeros are dots and poles are open circles. Here H_t, $0 \le t \le 2$, deforms the identity ($t = 0$) into F ($t = 2$) and runs the steps in turn ($H_t = H^{(k)}_{t-k+1}$ on $k-1 \le t \le k$): for f_1, the zoom homotopy $H^{(1)}_s(z) = (e^{sz} - 1)/s + s$ for $0 < s \le 1$, $H^{(1)}_0(z) = z$; for f_2, the elliptic Möbius transformations $H^{(2)}_s$ that fix $p = i$ and $q = -i$, with $H^{(2)'}_s(p) = e^{-\pi i s/2}$, where

$$F(z) = \frac{e^z - 1}{e^z + 1}, \qquad F = f_2 \circ f_1,$$
$$f_1(z) = e^z, \qquad f_2(w_1) = \frac{w_1 - 1}{w_1 + 1}.$$

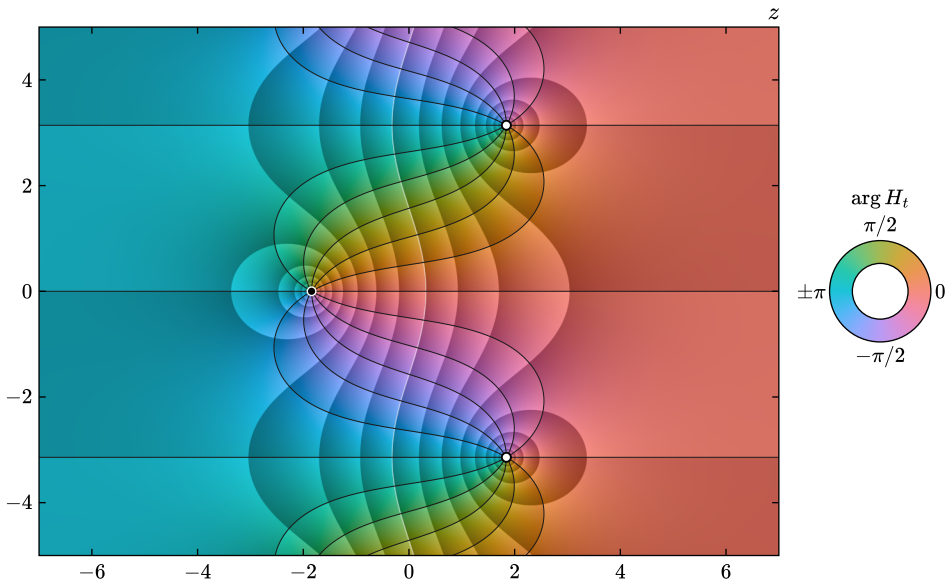


FIGURE 2. Phase portrait of H_t on $[-7,7]\times[-5,5]$: the color shows $\arg H_t$ (key at right), the thin lines are the level sets $\arg H_t \in \frac{\pi}{6}\mathbb{Z}$, and the shading steps in $|H_t|$ by factors of $e^{2\pi/12} \approx 1.69$; the white line is $|H_t| = 1$. Zeros (dots) at -1.843. Poles (open circles) at $1.843 + \pi i$ and $1.843 - \pi i$. Shown at $t = 1.20$. Here H_t, $0 \le t \le 2$, deforms the identity ($t = 0$) into F ($t = 2$) and runs the steps in turn ($H_t = H^{(k)}_{t-k+1}$ on $k-1 \le t \le k$): for f_1, the zoom homotopy $H^{(1)}_s(z) = (e^{sz} - 1)/s + s$ for $0 < s \le 1$, $H^{(1)}_0(z) = z$; for f_2, the elliptic Möbius transformations $H^{(2)}_s$ that fix $p = i$ and $q = -i$, with $H^{(2)\prime}_s(p) = e^{-\pi i s/2}$, where

$$F(z) = \frac{e^z - 1}{e^z + 1}, \qquad F = f_2 \circ f_1,$$
$$f_1(z) = e^z, \qquad f_2(w_1) = \frac{w_1 - 1}{w_1 + 1}.$$

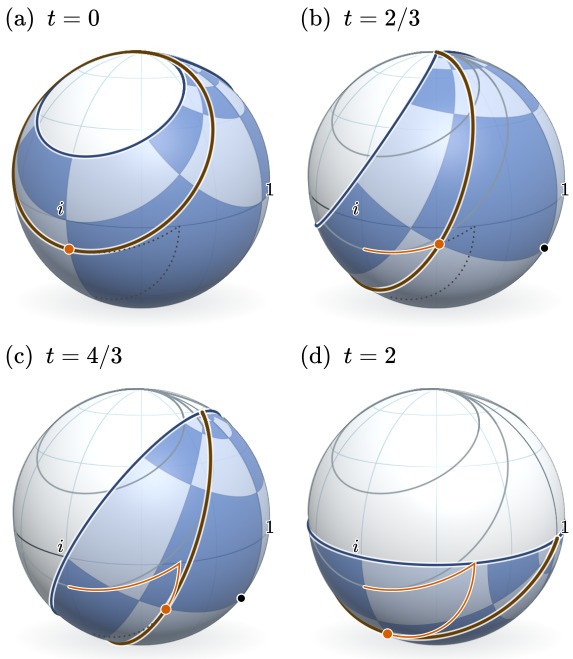


FIGURE 3. The deformation H_t on the Riemann sphere, with ∞ at the north pole, from the identity ($t = 0$) to F ($t = 2$), at (a) $t = 0$, (b) $t = 2/3$, (c) $t = 4/3$, and (d) $t = 2$: the image of the set $|\operatorname{Im} z| < \pi/2$, the line $\operatorname{Im} z = 0.8$, and the point $z_0 = 0$ under H_t. Solid: the path $t \mapsto H_t(z)$ of the point $0.8i$ (dot) up to the frame's t; dotted: the rest of the path; gray: the boundary at $t = 0$. The map is

$$F(z) = \frac{e^z - 1}{e^z + 1}.$$



FIGURE 4. The deformation H_t on the Riemann sphere, with ∞ at the north pole: the image of the set $|\operatorname{Im} z| < \pi/2$, the line $\operatorname{Im} z = 0.8$, and the point $z_0 = 0$ under F ($t = 2$) over its boundary at $t = 0$ (gray). The arrow is the path $t \mapsto H_t(z)$ of the point $0.8i$, from z (dot) to $F(z)$. The point $z_0 = 0$ is on the far side of the sphere and is not shown. The map is

$$F(z) = \frac{e^z - 1}{e^z + 1}.$$

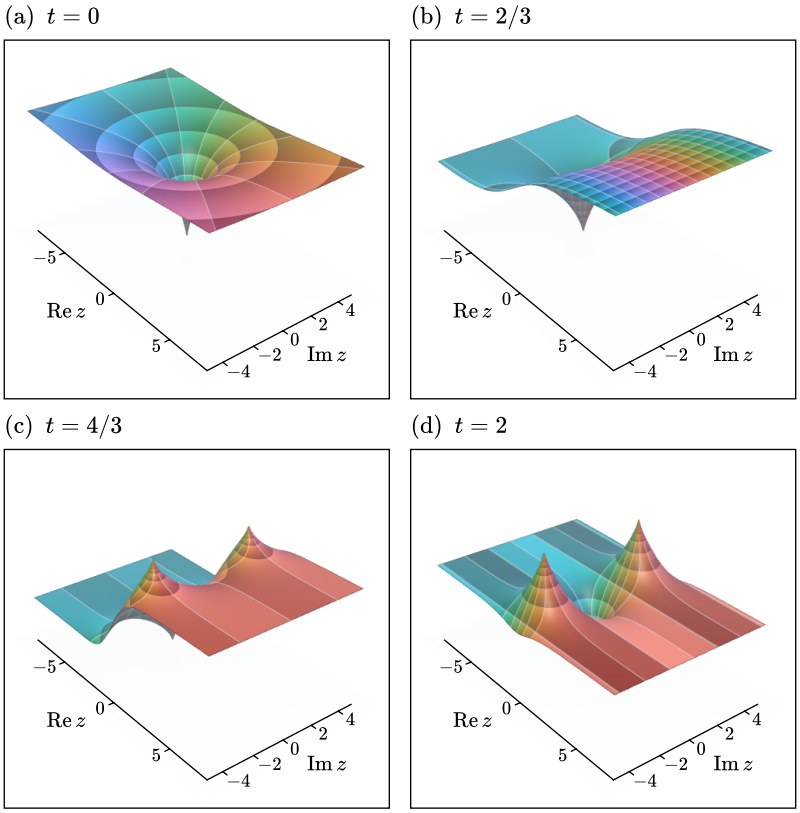


FIGURE 5. Landscapes of H_t over $[-8, 8] \times [-5, 5]$ at (a) $t = 0$, (b) $t = 2/3$, (c) $t = 4/3$, and (d) $t = 2$: the height is $\log|H_t|$ and the color shows $\arg H_t$; zeros are pits and poles are peaks. Here H_t, $0 \le t \le 2$, deforms the identity ($t = 0$) into F ($t = 2$) and runs the steps in turn ($H_t = H^{(k)}_{t-k+1}$ on $k - 1 \le t \le k$): for f_1, the zoom homotopy $H^{(1)}_s(z) = (e^{sz} - 1)/s + s$ for $0 < s \le 1$, $H^{(1)}_0(z) = z$; for f_2, the elliptic Möbius transformations $H^{(2)}_s$ that fix $p = i$ and $q = -i$, with $H^{(2)\prime}_s(p) = e^{-\pi i s/2}$, where

$$F(z) = \frac{e^z - 1}{e^z + 1}, \qquad F = f_2 \circ f_1,$$
$$f_1(z) = e^z, \qquad f_2(w_1) = \frac{w_1 - 1}{w_1 + 1}.$$

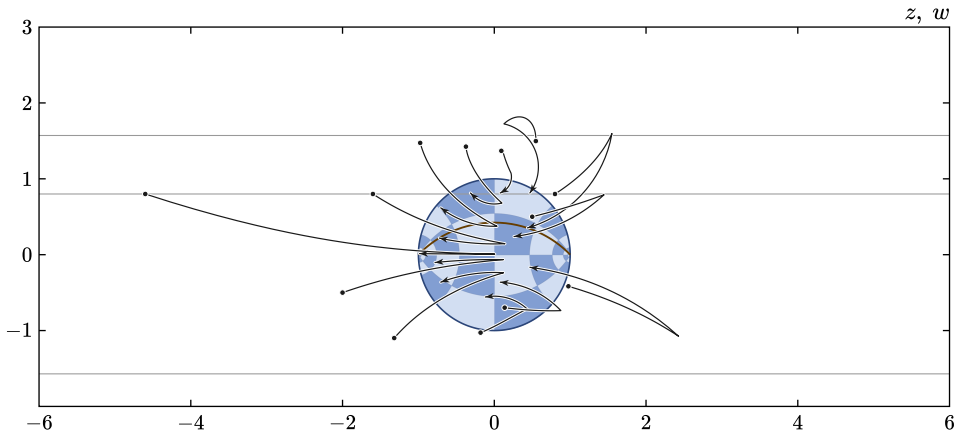


FIGURE 6. The set $|\operatorname{Im} z| < \pi/2$ and the line $\operatorname{Im} z = 0.8$ (gray outline), and their image under F (checkerboard), joined by the homotopy H_t, $0 \le t \le 2$, that runs the steps in turn ($H_t = H^{(k)}_{t-k+1}$ on $k-1 \le t \le k$): for f_1, the zoom homotopy $H^{(1)}_s(z) = (e^{sz} - 1)/s + s$ for $0 < s \le 1$, $H^{(1)}_0(z) = z$; for f_2, the elliptic Möbius transformations $H^{(2)}_s$ that fix $p = i$ and $q = -i$, with $H^{(2)\prime}_s(p) = e^{-\pi i s/2}$, where

$$F(z) = \frac{e^z - 1}{e^z + 1}, \qquad F = f_2 \circ f_1,$$
$$f_1(z) = e^z, \qquad f_2(w_1) = \frac{w_1 - 1}{w_1 + 1}.$$

Arrows are the paths $t \mapsto H_t(z)$ of 3 boundary points and 10 interior points, from z (dot) to $F(z)$. Regions are shaded by a checkerboard of squares of side 1 in the z-plane, carried by the map.

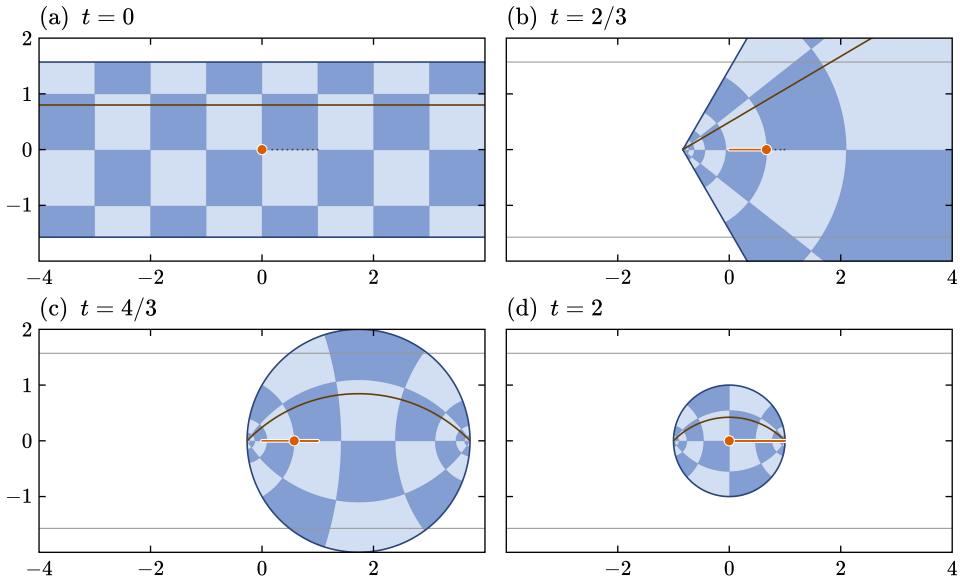


FIGURE 7. The deformation H_t from the identity ($t = 0$) to F ($t = 2$), at (a) $t = 0$, (b) $t = 2/3$, (c) $t = 4/3$, and (d) $t = 2$, where

$$F(z) = \frac{e^z - 1}{e^z + 1}.$$

Solid: the paths $t \mapsto H_t(z)$ of the marked points up to the frame's t; dotted: the rest of each path; gray: the boundary at $t = 0$.